

RIZQI INDIRSYAH FADHIL

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Greater Bandung, Indonesia

Graduate Mechanical Engineering student with experience in design, structural, thermal, Vibration analysis and maintenance, Microcontroller, Sensor, and Instrumentation Projects. Proficient in CAD modeling, Mechatronics, simulation-based evaluation, and standards-driven technical assessment. Strong analytical, problem solving, collaborative and leadership skills.

Work Experiences

PT Gexcon indonesia - Bandung

July 2024 - Dec 2024

Engineer Intern

- Providing technical support to the Engineering team in geometry modeling, CFD simulations, and proposal preparation
- Working with engineering teams to solve problems with CFD simulations set up using FLACS software, and building several geometry needed for projects.
- Collaborating with multidisciplinary teams and maintaining strong client relationships across multiple international company on risk assesment such as PT Kilang Pertamina Internasional Unit IV Balongan, PT Tripatra, and Gexcon Australia

PUSTEKHAN ITB - PT Robo Marine Indonesia – Bandung

July 2024 - November 2024

Research Engineer Intern

- Conducting several simulation of thrust and torsion using Ansys Fluent for ROV (Remotely Operated Underwater Vehicle)'s propulsion and validating the data using experimental testing.
- Optimizing the propeller using toroidal blade and minimizing the clearance of the propeller's blade to improve propeller thrust and torsional performance.

PT Supreme Energy Muara Laboh – Solok Selatan

Jan 2024 - Feb 2024

Engineer Intern

- Learning and Assisting in the design and analysis of mechanical components and systems for geothermal power generation, including pipes, turbines, pumps, heat exchangers, cooling tower.
- Participate in the implementation of preventive maintenance programs to ensure optimal performance and reliability of mechanical equipment and cooling tower.
- Conduct field inspections and assessments to identify opportunities for efficiency improvements and cost savings in pipe structure.

Education & Academic Experiences

Institut Teknologi Bandung - Bandung

Aug 2021 – March 2026

Bachelor of Mechanical Engineering

- Theses: DESIGN, FABRICATION, AND TESTING OF AN ELECTRONIC DEVICE THAT GENERATES SIMULATED KEYPHASOR SIGNAL FOR INTERMEDIATE SHAFT
- GPA: 3.40/4.00
- Course Assistant & Lecturer's Assistant of MS2111 Kinematics and Dynamics of Machinery on 5th semester

Dynamic Laboratory ITB

Feb 2025 – March 2026

Laboratory Assistant

- Helped and taught students with theory and hands-on practice with several equipments and method used in industrial such as balancing and alignment, electrical circuit, pneumatic system and PLC demonstration.
- Assisted with several practicum that were conducted in dynamics laboratory of ITB. The practicum consisted of 2 course, which were Basic Mechatronics (MS2204) and Mechanical Maintenance (MS4001)

Basic Physic Laboratory ITB

Feb 2023 – April 2023

Laboratory Assistant

- Helped catalyze the understanding of Basic Physics Course (FI1101 & FI1102) for firstyear students at ITB through direct application of the theories learned in class in the form of Basic Physics practicums, especially in the field of mechanics
- Assisted with 6 Basic Physics lab practicum and taught 30 practitioners how to use kits and assemble practicum experiments.

Organizational Experiences

Himpunan Mahasiswa Mesin ITB

Nov 2023 - May 2024

Vice Head of Competition Division

- Co-led a 10-member team in re-establishing the annual competition after a long inactive period.
- Developed event framework, timeline, task delegation, and budget planning to support future implementation.

BPA FTMD 2021

Dec 2021 – April 2022

Academic Divison Staff

- creating an academic framework for FTMD 2021 and teaching lesson as tutor for Pengkom FTMD 2021.
- collaborate with others to prepare any necessary peer tutor, notes, and recording.

Projects and Skills

- **Projects:** PKM (Pekan Kreativitas Mahasiswa) fire tank, ROV (Remotely Operated Vehicle), Soapbox RCKT, IoT.
- **Soft Skills:** Analytical problem solving, , project coordinating, leadership, technical reporting
- **Technical Skills :** Simulation/Analysis (Ansys Structural, Ansys Fluent), CAD (SolidWorks 2D/3D, AutoCAD 2D), Engineering Standard, PCB (Kicad), 3D Print (Ultimaker Cura, Prusa slicer), Programming/Data (Python, Matlab, Excel), Microcontroller (ESP32 and Arduino with Arduino IDE)